



DO-IT-YOURSELF: PROJECT

built in, so you do not need to install Python separately. Just install Thonny in your computer and you are ready to go!

Once you have installed Thonny IDE, connect your Raspberry Pi Pico board with your computer through an appropriate USB data cable. You need to press and hold the BOOT-SEL button on the Raspberry Pi Pico board (B1) while connecting it to your computer using a USB-A to

micro-USB data cable.

After successfully connecting to PC, you can see your Raspberry Pi Pico as USB mass storage device.

Now you can open Thonny IDE and there will be a popup that shows some information regarding Raspberry Pi Pico. Just proceed to install the MicroPython firmware. Your Raspberry Pi Pico board is now updated with MicroPython firmware and is ready to execute Python codes.

(Actually, there is also a drag and drop way to do this. Check <https://www.twilio.com/blog/programming-raspberry-pi-pico-micro-controller-micropython>)

Copy-paste the above Python code in Thonny editor window (Fig. 4) and save the code before you run it. See there are two floppy disc buttons—open and save.

Choose Raspberry Pi Pico in the “where to save to” popup window, as shown in Fig. 5, assign a name for your Python file with an extension ‘py’ and click OK.

Once the code is saved, the green Play button on top can be used to execute the code in Raspberry Pi Pico. To stop the code’s execution, use the red Stop button.

Remember, the noise generation code should be run automatically

EFY Note

The source code of this project is available for free download at source.efymag.com

to power the standalone Raspberry Pi Pico setup. So, when saving the code file on the Raspberry Pi Pico board, name it *main.py* (Fig. 6).

Awesome? Well, just do it your own way!

Testing

Testing of this project is simple if you know what you are going to do! After successful construction, just run the prototype with a USB power bank or a similar USB power source. If all goes well, the tone signal would be heard. The pushbuttons S1 and S2 can be used to increase or decrease the tone frequency.

The author has been playing with Raspberry Pi Pico a lot lately. He has written on some online channels about Raspberry Pi Pico, which is an awesome minuscule microcontroller board capable of running code to control physical hardware. Pin details of the Pico board are shown in Fig. 7.

The Raspberry Pi Pico used for this project already had headers soldered on, and the author had previously put MicroPython on it. He is using Thonny 3.3.3 (Fig. 8) on his Windows 10 (64-bit) computer.

The author’s prototype wired on the breadboard is shown in Fig. 9.

This tutorial is presented by the author for you to try and learn. You may improve the code for better application. You may even take this further into the realm of professional electronic tone generators. It is hoped you will find more interesting ways to play with audio on the Raspberry Pi Pico! **EFY**



Fig. 8: Thonny IDE

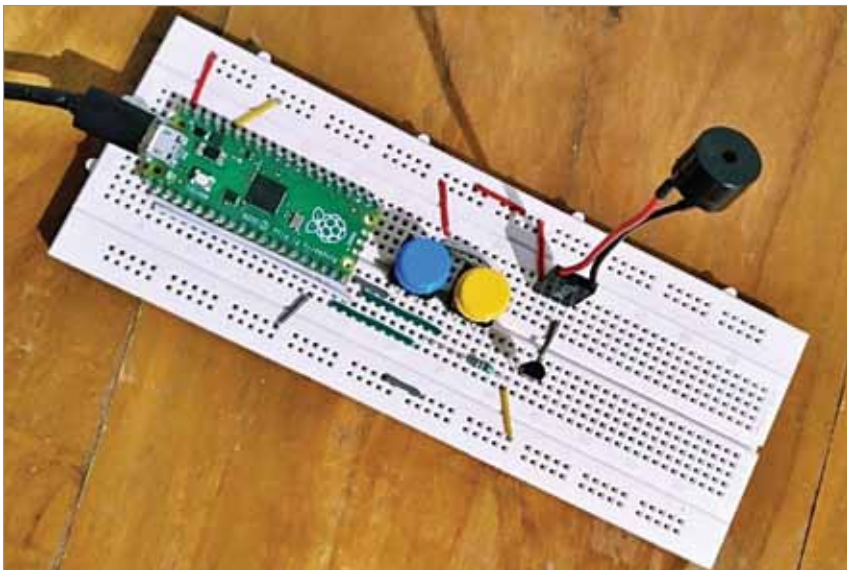


Fig. 9: Author's prototype on breadboard



T.K. Hareendran is an electronics designer, hardware beta tester, technical author, and product reviewer